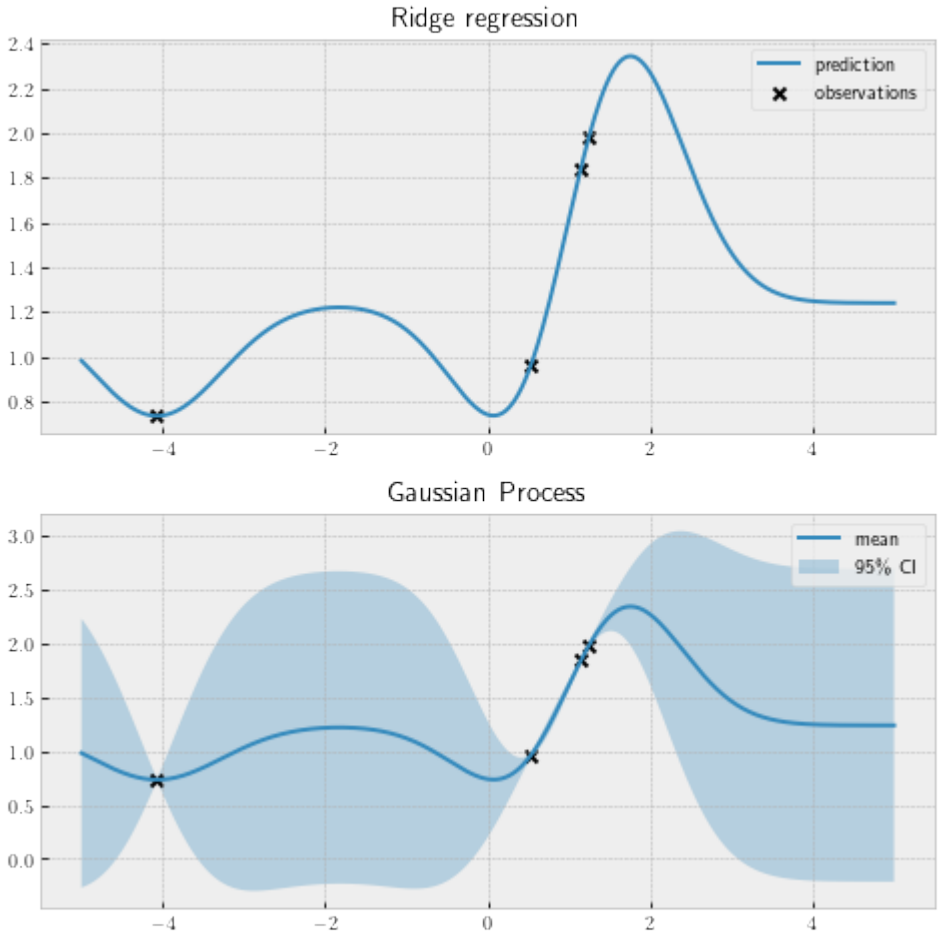


Figure 1.4 Non-Bayesian models such as a ridge regressor makes point-wise estimate while a GP produces probability distributions as predictions. The GP thus offers calibrated quantification of uncertainty, an important factor when making high-risk decisions.

We first fit a ridge regression model on this data set and make predictions in the range between -5 and 5. The top panel of Figure 1.4 shows these predictions. Each prediction made by this model at a given test point is a single-valued number, which does not capture our level of uncertainty about how the function we are learning from behaves. For example, given the test point at 2, this model simply predicts 2.2.

A ridge regression model is a modified version of a linear regression model, where the weights of the model are regularized so that small values are preferred.

We don't need to go into too much detail about the inner workings of this model. The point here is that a ridge regressor only produce point estimates without a measure of uncertainty, which is also the case for many machine learning models such as support-vector machines, decision trees, and neural networks.

How, then, does a GP make its predictions? As shown on the bottom panel of Figure 1.4, predictions by a GP are in the form of probability distributions (specifically normal distributions). This means at each test point, we have a mean prediction (the solid line) as well as

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